i. Air friction damping

Figure 6.3 shows an arrangement for obtaining air friction damping. This arrangement consists of a light aluminium piston which is attached to the moving system. The piston moves in a fixed air chamber which is closed at one end. The clearance between piston and wall chamber is uniform and small.

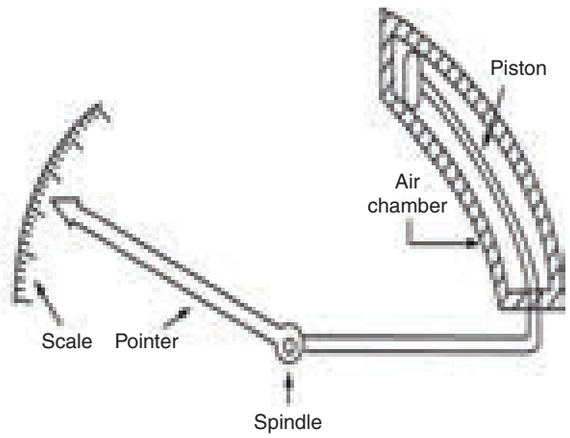


Fig. 6.3 Air friction damping

When the piston moves into the chamber, air inside the chamber is compressed and thus the pressure inside the chamber opposes the motion of the piston and hence whole of the moving system. When the piston moves out of the chamber, the pressure inside the chamber falls and the pressure on open side becomes greater than on the other side and thus there is again an opposition to motion of the piston.

ii. Fluid friction damping

This method is similar to air friction damping. In this only air is replaced by damping oil. The damping oil employed must have the following properties:

i. Must be a good insulator.
ii. Viscosity of oil should not change with temperature.
iii. Should be non-evaporating.
iv. Should be non-corrosive.

In this method, a disc attached to the spindle of moving system is immersed in damping oil as shown in figure 6.4. When the moving system moves, the disc moves in oil and a frictional drag is produced which opposes the motion of the moving system. Damping force due to fluid is greater than that of air due to more viscosity.

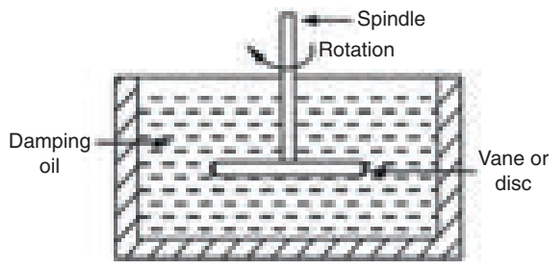


Fig. 6.4 Fluid friction damping

iii. Eddy current damping

In this method, an aluminium or copper disc is connected to the spindle. The disc is positioned between the poles of a permanent magnet as shown in figure 6.5. If the disc moves, it cuts the flux which causes an induced e.m.f. in the disc. As the disc is a closed path, induced emf circulates eddy currents in the disc.

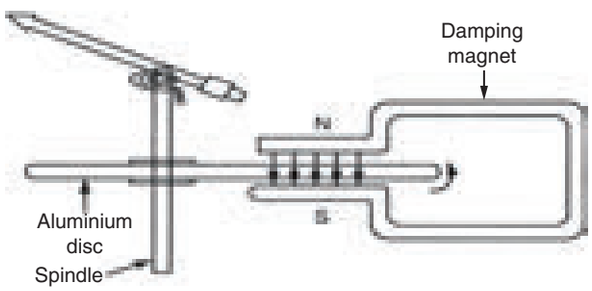


Fig. 6.5 Eddy current damping

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